It means that all the terms are arranged on the y-axis according to their serial numbers. As far as I know, such sequence are called $\boxed{\text{increasing}}$.

$\boxed{\text{Author:}}$ A more general case is that of $\boxed{\text{nondecreasing}}$ sequences provided we add the equality sign to the above series of inequalities.

$\boxed{\text{Definition:}}$ A sequence (y_n) is called nondecreasing if $y_1 \leq y_2 \leq y_3 \leq \ldots \leq y_n \leq \ldots$

A sequence (y_n) is called nonincreasing if $y_1 \geq y_2 \geq y_3 \geq \ldots \geq y_n \geq \ldots$

Nondecreasing and nonincreasing sequences come under the name of monotonic sequences.

Please, identify monotonic sequences among examples (1)–(13).

$\boxed{\text{Son:}}$ Sequences (1) to (4) $\longrightarrow$ monotonically increasing sequence

Sequences (6), (7) $\longrightarrow$ monotonically decreasing sequence

Sequences (8), (9) and (10) are not monotonic.

$\boxed{\text{Author:}}$ $\boxed{\text{Definition:}}$ A sequence (y_n) is $\boxed{\text{bounded}}$ if there are two numbers A and B labelling the range which encloses all the terms of a sequence.

$A \leq y_n \leq B \quad (n = 1, 2, 3, \ldots)$

If it is impossible to identify such two numbers (or, in particular, one can find only one of the two such numbers, either the least or the greatest), such a sequence is $\boxed{\text{unbounded}}$.

Do you find bounded sequences among our examples?

$\boxed{\text{Son:}}$ Sequence (5) and (7) are bounded.

$\boxed{\text{Author:}}$ Find the numbers A and B for sequence (5)

$\boxed{\text{Son:}}$ $A = 0$ and $B = 1$

$\boxed{\text{Reader:}}$ $A = \frac{1}{2}$ and $B = 1$

$\boxed{\text{Author:}}$ Of course, but if there exists even one pair of A and B, one may find $\boxed{\text{any number}}$ of such pairs. You could say, for example, that $A = 0, B = 2$, (Imp point)

or $A = -100, B = 100$, etc. and be equally right.

$\boxed{\text{Reader:}}$ Yes, but my numbers are more accurate.